

Antonio Leblanc

Co-Founder & CTO — 1.5°C
AI Wildfire Platform

Rio de Janeiro, Brazil · French & Brazilian citizen
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190 cameras 150 HaaS on hardware we operate + 40 SaaS on third-party cameras	40M+ hectares Monitored across Pantera's full platform — camera, multi-satellite, and geospatial data	9 AI agents Self-hosted Hermes fleet — 426 merged PRs, 83% human acceptance, 7 repos	10 months Intern to CTO, 2020 — thrown into the role when the founding engineer left
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EXPERIENCE

Co-Founder & CTO — 1.5°C, umgrauemeio Sep 2020 – Present
Rio de Janeiro, Brazil

- Lead Pantera end to end, architecting the hybrid edge/cloud platform behind 190 field cameras — integrating camera, multi-satellite, and geospatial data across risk analysis, early detection, fire-spread modeling, CO₂ estimation, and deforestation monitoring — **tuned for recall**, since a missed fire costs more than a false alarm.
- Built the engineering, ML, and environmental science team — **hired and developed every person on it**; still review code and architect alongside the team.
- Deployed **9 AI agents** on self-hosted **Hermes** (decoupled inference) — built an issue-to-PR pipeline with **426 merged PRs** across 7 repos (**83%** human acceptance, ~70% code & infra, 2-PR limit) and **158 issues closed**; cut cloud costs **15%**.
- Built and handed off **Customer Success, Technical Support, and Marketing** functions to dedicated owners as each outgrew founder-level involvement.
- Technical lead on the enterprise deal and deployment of **Suzano, the world's largest pulp producer** and our largest customer, won in a competitive bid — plus strategic-partner and investor presentations driving commercial expansion into agriculture and conservation.

Software Engineer Apr – Aug 2020
Machine Learning Intern Nov 2019 – Mar 2020

Data Science Intern — Radix Engenharia e Software Jun 2019 – Oct 2019
Rio de Janeiro, Brazil

- Built machine learning models for predictive maintenance on industrial equipment — time-series, digital twin.
- Developed a genetic optimization algorithm that reduced energy consumption in production processes by 20%.

Operations Management Intern — Amazon Apr 2018 – Oct 2018
Orléans, France

- Designed and implemented supply chain optimization projects in a fulfillment center processing 400K units/day.
- Managed a team of 20 associates to streamline operations and enhance productivity.

OPEN SOURCE

ForeFire — open-source wildfire spread simulator in C++, developed at CNRS / Université de Corse 2022 – Present

- Contributor since 2022, maintainer since 2025 — reviewing and merging PRs, cutting releases, Dockerization, CI/CD, documentation.
- Co-author on the project's peer-reviewed **JOSS paper** (DOI 10.21105/joss.08680).

EDUCATION

General Engineering 2016 – 2018
École Centrale de Lyon, France

Mechanical Engineering 2014 – 2019
Polytechnic School of UFRJ, Brazil ·
CAPES scholarship for academic excellence

Baccalauréat Scientifique 2013
Lycée Molière, Rio de Janeiro, Brazil · Highest Honors

SKILLS & LANGUAGES

LANGUAGES	Native: Portuguese, French · Fluent: English · Professional: Spanish
STACK	Python, FastAPI, TypeScript/Angular, AWS, Docker, PostGIS, PyTorch
ML/AI	Deep learning, geospatial ML in production, LLMs and agentic systems
LEADERSHIP	Agile/OKRs, cross-functional teams across engineering, ML, and environmental science
ALSO	Marathon (3:23) · 10 years of formal piano; self-taught saxophone, guitar